Chapter 7 | Data Handling — Pandas / Data Science & Big Data

Complete chapter notes — all concepts of the syllabus in teaching order. Points marked ✦ at the end of each section are the facts asked in previous exam papers (DSSSB, KVS, NVS, HTET, BPSC, RPSC etc.).

7.1 Data Science and Big Data

Data science: the discipline of extracting knowledge and insight from data — combining statistics, programming and domain understanding. Its standard working pipeline:

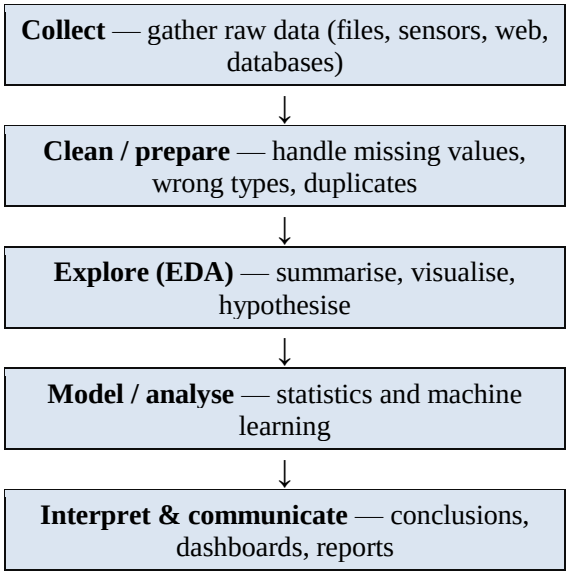


Fig. 7.1 — The data-science pipeline. The exam's entry points are the EDA and analysis stages.

(1) Exploratory Data Analysis (EDA): its tested advantage is that it offers **flexible ways to GENERATE HYPOTHESES** from the data before any formal testing — it is exploratory precisely because it does not aim at definitive answers.

(2) Growth curve analysis is the **specialised set of techniques for modelling CHANGE OVER TIME**, and it doubles as a **data-reduction technique** (many observations summarised by a few curve parameters).

(3) Big Data is data too large or fast for traditional tools, characterised by the **V's**: Volume (size), Velocity (speed of arrival), Variety (forms), Veracity (trustworthiness), Value. **Hadoop** is the classic open-source framework: **HDFS** (distributed storage across cluster nodes) + **MapReduce** (distributed processing — map splits the work, reduce combines the results); Spark is its faster in-memory successor.

(4) Data warehouse: **a relational database designed for QUERIES AND ANALYSIS rather than transaction processing** (the tested definition) — it integrates historical data from many sources (OLAP), while the operational database handles day-to-day transactions (OLTP).

Note: One project-management stray is keyed here: **Communication, Schedules and Task List are key functional aspects of PM software**, and **software integration is a valid selection criterion** — all three statements true alongside the warehouse definition.

Exam Points — 7.1

✦ EDA → flexible hypothesis generation. Growth curve analysis = change over time + data reduction.

✦ Data warehouse = relational DB for queries/analysis, not transactions. PM software: Communication + Schedules + Task List; integration is a selection criterion (all true).

7.2 Pandas — Series, DataFrame and Operations

Pandas is Python's data-handling library, built on NumPy, imported as **import pandas as pd**. If that line raises an **ImportError, it means Python could NOT find Pandas in its list of available libraries** — only statement III; the fix is pip install pandas.

Structure	Identity (both directly asked)
Series	**A ONE-dimensional labelled array capable of holding ANY data type** — integers, strings, floats, Python objects; the labels form its index
DataFrame	**A TWO-dimensional labelled data structure with columns of potentially DIFFERENT types** — rows × typed columns, like a spreadsheet or SQL table; each column is itself a Series

(1) Creation: pd.Series([10, 20, 30]) — default index 0, 1, 2; pd.DataFrame(dict-of-lists) — keys become columns; pd.read_csv("file.csv") loads a table, df.to_csv() saves one.

(2) **set_index() takes a column name (or a list of names) and creates a new, re-indexed DataFrame** — the asked re-indexing method (reset_index() undoes it). head()/tail() preview rows; df.shape gives (rows, columns); df.describe() summarises the statistics of 7.3.

(3) **MERGE is the operation that concatenates multiple tables both column-wise and row-wise as a database does** — pandas' SQL-style join on key columns (concat() merely stacks frames along an axis; groupby() aggregates by group; these mirror SQL's JOIN and GROUP BY).

(4) **Missing data** appears as NaN: dropna() removes incomplete rows, fillna(value) substitutes; isnull() tests. Missing data, iteration and these operations are the distractor options of the Series/DataFrame questions — keep the definitions separate.

Exam Points — 7.2

✦ ImportError → Python couldn't find Pandas (only III). Series = 1-D labelled array, any type; DataFrame = 2-D labelled, differently-typed columns.

✦ set_index() = column-based re-indexing into a new DataFrame. merge = table joining column- and row-wise (database style).

7.3 Descriptive Statistics for Data Handling

Data analysis rests on a handful of formulas — pandas computes each of them (mean(), median(), mode(), var(), std(), describe()), and the exams expect the formulas themselves. For n observations $x_1, x_2, \ldots, x_n$:

Mean (average) — the balance point of the data:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Median — the middle value of the sorted data: the ½(n + 1)-th value for odd n, the mean of the two central values for even n; **mode** = the most frequent value. The median resists outliers; the mean does not.

Range and variance — how spread out the data is. Range = maximum − minimum; variance averages the squared deviations from the mean:

$$\sigma^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n}$$

Standard deviation — the square root of the variance, back in the data's own units (pandas' std() uses n − 1, the sample form):

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n}}$$

Normalisation — preparing columns for comparison or modelling. Min-max scaling squeezes values into [0, 1]; the z-score states how many standard deviations a value sits from the mean:

$$x' = \frac{x - min}{max - min} \qquad z = \frac{x - \bar{x}}{\sigma}$$

Worked: Data 2, 4, 6, 8, 10: mean = 30/5 = **6**; median = **6**; range = **8**; deviations −4,−2,0,2,4 → squares 16,4,0,4,16 → variance = 40/5 = **8**; σ = √8 ≈ **2.83**; z-score of 10 = (10−6)/2.83 ≈ **1.41**; min-max of 8 = (8−2)/(10−2) = **0.75**.

Exam Points — 7.3

- ✦ Mean = Σx/n; median = middle of sorted data (outlier-resistant); mode = most frequent; range = max − min.
- ✦ Variance = mean of squared deviations; σ = its square root; z = (x − mean)/σ; min-max maps to [0, 1]. pandas: mean/median/mode/var/std/describe.

Quick Revision — 12 points

1. Pipeline: collect → clean → explore → model → interpret.
2. EDA's advantage = flexible hypothesis generation (not definitive answers).
3. Growth curve analysis = modelling change over time + data reduction.
4. Big Data V's: Volume, Velocity, Variety, Veracity, Value. Hadoop = HDFS (storage) + MapReduce (processing).
5. Data warehouse = relational database for queries and analysis, NOT transaction processing (OLAP vs OLTP).
6. PM software: Communication, Schedules, Task List; software integration = selection criterion (all true).
7. import pandas as pd; ImportError = Python could not find Pandas (statement III only).
8. Series = 1-D labelled array holding any data type; DataFrame = 2-D labelled structure, columns of different types; each column is a Series.
9. set_index(column or list) → new re-indexed DataFrame; head/tail preview; describe() summarises.
10. merge = database-style joining of tables column- and row-wise; concat stacks; groupby aggregates; NaN handled by dropna/fillna.
11. Mean = Σx/n; median = sorted middle; mode = most frequent; range = max − min; variance = mean squared deviation; σ = √variance.
12. z = (x − mean)/σ; min-max x′ = (x − min)/(max − min) → [0, 1]. Worked set 2,4,6,8,10: mean 6, variance 8, σ ≈ 2.83.